

Discussion 1 Posts

Initial Post

Cybersecurity as a Global Issue

Advances in technology have allowed the world to become more interconnected, giving businesses a global profile. This has many advantages, such as a larger pool of customers and vendors, and quick access to company data. It also creates problems, such as greater exposure to cyber-attacks, and adhering to differing data storage laws.

Importance of Investing in Cybersecurity

Cybercrime has risen alongside the development of technology. It now makes up around 50% of crime, and companies must worry about multiple types of attackers (Anderson 2020). In the UK, around half of businesses reported a cyber-attack during 2023, with the numbers being higher as businesses got bigger; around three quarters of large businesses were attacked (Department for Science, Innovation and Technology, 2024).

Some governments have tried to influence businesses to take a more serious approach to cybersecurity. In 2022 the UK government produced a new National Cyber Strategy; one of its 5 pillars focused on reducing cyber risks so businesses can get more from digitisation, whilst also making customers feel their data is more secure (Cabinet Office, 2022).

Security of data should be a business concern, as should where that data is sourced and stored. Different rules and regulations are in place depending on where those the data relate to is from. Failure to comply can lead to fines and reputational damage for companies and organisations.

Case Study: Marriott International, Inc.

Marriott International, Inc., a global hotel chain, was the subject of 3 data breaches between 2014 and 2020 at its subsidiary Starwood, that affected over 344 million customers (Federal Trade Commission, 2024). As a result, its reputation was damaged, leading to a drop in its share price (Volodzko, 2018), and fines from multiple regulatory bodies (Federal Trade Commission 2024).

Starwood became a subsidiary of Marriott International Inc. in 2016, with the first breach happening prior to the acquisition, showing that organisations must pay attention to cybersecurity in all areas, including potential subsidiaries, partners, and vendors.

Conclusion

Cybersecurity's importance increases globally as technology makes the world more interconnected. Businesses should invest in cybersecurity if they want to take advantage of this, as the benefits of being a global company are partnered with greater risks and need to follow an increasingly number of laws and regulations.

Word Count: 379

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Responses to Peers

Reply to Peer Response – Doug

Hi Doug, thanks for your response.

The issue of making software secure-by-design is one that has taken on greater prominence over the last few years, with the United States leadership calling for responsibility of digital security to be taken away from individuals and end-users of software and placed on technology providers (National Cybersecurity Strategy, 2023).

The prevailing paradigm in the industry is to push products and services out to market as soon as possible, adding security at a later point. This is based on the idea that getting your products into the domain quickly, by making them user friendly, and easy

for application developers to work with, gives a better chance at a dominant market position (Anderson, 2020). Although this is smart business practice, it does make these products and services intrinsically insecure. This same paradigm assigns much of the blame for successful cyber-attacks and data breaches onto the end-user, leaving the software designer at no risk of liability.

The status quo, and legal difficulties around defining how liability pertains to software (Hamin et al., 2023), have meant that legal challenges around insecure software are complex and rare. However, CrowdStrike and Delta Airlines are suing each other over liability for Delta's losses during a system outage caused by a faulty software update (Shepardson, 2024), a case that could give some clarity and accelerate change in this area.

In the case of the Marriott International Inc. data breaches, attackers, thought to be state-sponsored Chinese hackers, installed a Remote Access Trojan (RAT), alongside a username / password sniffer, having likely gained access to systems via a phishing attack (Fruhlinger, 2020). This gives greater importance to the necessity of secure-by-design, with the goal of limiting the potential of attacks taking place after some form of user mistake. In this case Marriott International Inc. were found responsible for the data breaches and were fined £18.4 million by ICO for violating GDPR regulations (ICO, 2020).

Word Count: 320

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Peer Response - Aisha

Although the global outage you have mentioned is not considered a cybersecurity event, it had the same effect as a DoS or DDoS attack, denying availability of services to millions. Airports should have provisions in place to respond to these situations, given that DDoS attacks are the most common type of attack on the aviation sector (Sysdream, 2024). Many airlines have experienced DDoS attacks on their websites in the recent past (Toulas, 2022), so they should be aware of their status as targets.

Most DoS attacks are motivated by political or activist ideals, and a significant number performed by Russian groups as retaliation to countries opposing the conflict in Ukraine (Enisa, 2023). Aviation is considered critical infrastructure (CISA, 2024), and airports and airlines in countries opposing Russian actions should expect to see continued DoS and DDoS attacks, and should prepare accordingly.

If airports have not already done so, implementing your suggestion of an alternative system that could be used in an emergency is a good idea. They should also create both an incident response plan and a disaster recovery plan, or update existing plans, with cybersecurity and DoS/DDoS attacks included. Implementing MFA and training staff to better understand cybersecurity are both effective suggestions to improve general security.

Word Count: 207

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Peer Response - Precious

I agree with your concerns about IoT devices and their impact on the cybersecurity field. The number of cyber-attacks on IoT devices has climbed exponentially in recent years, from around 32 million in 2018, to more than 112 million in 2022 (Petrosyan, 2023).

Your point that weak security in IoT devices has led to this rise is also valid. Many IoT devices are intrinsically weak; developers often care more about making the device easy to mass produce and user friendly, than addressing security concerns. This can lead to their firmware being lightweight and not powerful enough to offer strong security (TrendMicro, 2020). This is compounded by frequent use of default credentials in IoT devices that can be obtained by attackers (Shah, 2019). Long term care is also lacking; many of the devices are unlikely to receive any patches in their lifecycle (Anderson, 2020). Given your point about the rise in zero-day attacks, this is particularly worrying.

These vulnerabilities make IoT devices a target for use in botnets that launch large-scale internet attacks, with Mirai botnets being the most famous example. Mirai is malware that exploits default credentials to access IoT devices and add them to a network of infected devices, often to be used in DDoS attacks (Buxton, 2022). One of these attacks was on the company Dyn, and led to what some at the time considered the biggest in history, having brought down sites like Netflix, Twitter, Reddit and many other US and European sites (Woolf, 2016).

Word Count: 248

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Peer Response - Musthaq

Cathay Pacific Data Breach

The Cathay Pacific Airways data leaks 2014 - 2018 are another example of your point that corporations face substantial risk if they are not prepared to deal with cyber threats. Data of over 9 million customers were exposed in what Cathay Pacific announced as a brute force password attack (BBC, 2020). The ICO found several issues, including obsolete and unpatched systems, and deemed the overall cyber security of the corporation weak (Cowley, 2020).

This breach had an impact on the reputation and finances of the company. The day after disclosing the data breach, Cathay Pacific's stock price dropped to its lowest point in 9 years (Roantree, 2018). It also received a fine of £500,000 from the ICO for breaking the Data Protection Act 1988 (ICO, 2020).

Overall, this case strengthens your position, whilst also highlighting that ignoring cybersecurity and allowing systems to become outdated shows a lack of understanding of the role cybersecurity plays in running a successful corporation.

The Aviation Industry

Both the British Airways and Cathay Pacific data breaches point to the aviation industry being a target for cybercriminals looking to attack data confidentiality. However, 89% of attacks in 2023 were DDOS attacks, an attack on availability (Sysdream, 2024).

These figures suggest that the main reason for cyber attacks on the aviation industry today is to lower the availability of services, rather than to breach confidentiality. Accordingly, the industry should look at cybersecurity solutions beyond just those for data protection.

Word Count: 246

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Summary Post

My initial post looked at how advancements in technology have made the world more interconnected, something which has brought both positives and negatives to businesses. One negative has been greater exposure to cyber-attacks, which businesses report dealing with frequently (Cabinet Office, 2022).

The growing threat and impact of these attacks has compelled governments to develop policies and strategies to help businesses become more cybersecure. This led to discussion that government regulation may adversely affect innovation (Alshamsi, 2024) but this is balanced by research showing that cybersecurity risk has a bigger impact, with companies putting research and development funds towards precautionary funds instead (Wang et al. 2024).

I concluded in my initial post that companies should pay attention to how they secure data at every level, as failure to do so could lead to fines and reputational damage, with a data breach at Marriott International Inc analysed as a case study. A point was made about whether there are more serious consequences to successful cyber-attacks, with hospital service disruption, and national security being mentioned (Neary, 2024).

With cybersecurity a global issue it stands to reason that attacks on international companies and the data of citizens from various countries can be seen as a national security threat; the perpetrators of the Marriott breach are thought to have been state-sponsored Chinese nationals looking to gather data on American nationals (Fruhlinger,

2020). For disruptive attacks on critical infrastructure, we can look to the aviation industry, where DoS attacks are common (Sysdream, 2024).

A further discussion was about who is responsible for cybersecurity issues more generally, and why secure-by-design is not emphasised (Milward, 2024). Secure-by-design has started to gain traction at governmental level recently, with leaders calling for the burden of security to be taken away from end users (The White House, 2023). Currently security is seen as a feature (Deogun et al. 2019), something to add on later after rushing products to market to get a dominant market position (Anderson, 2020).

Word Count: 327

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